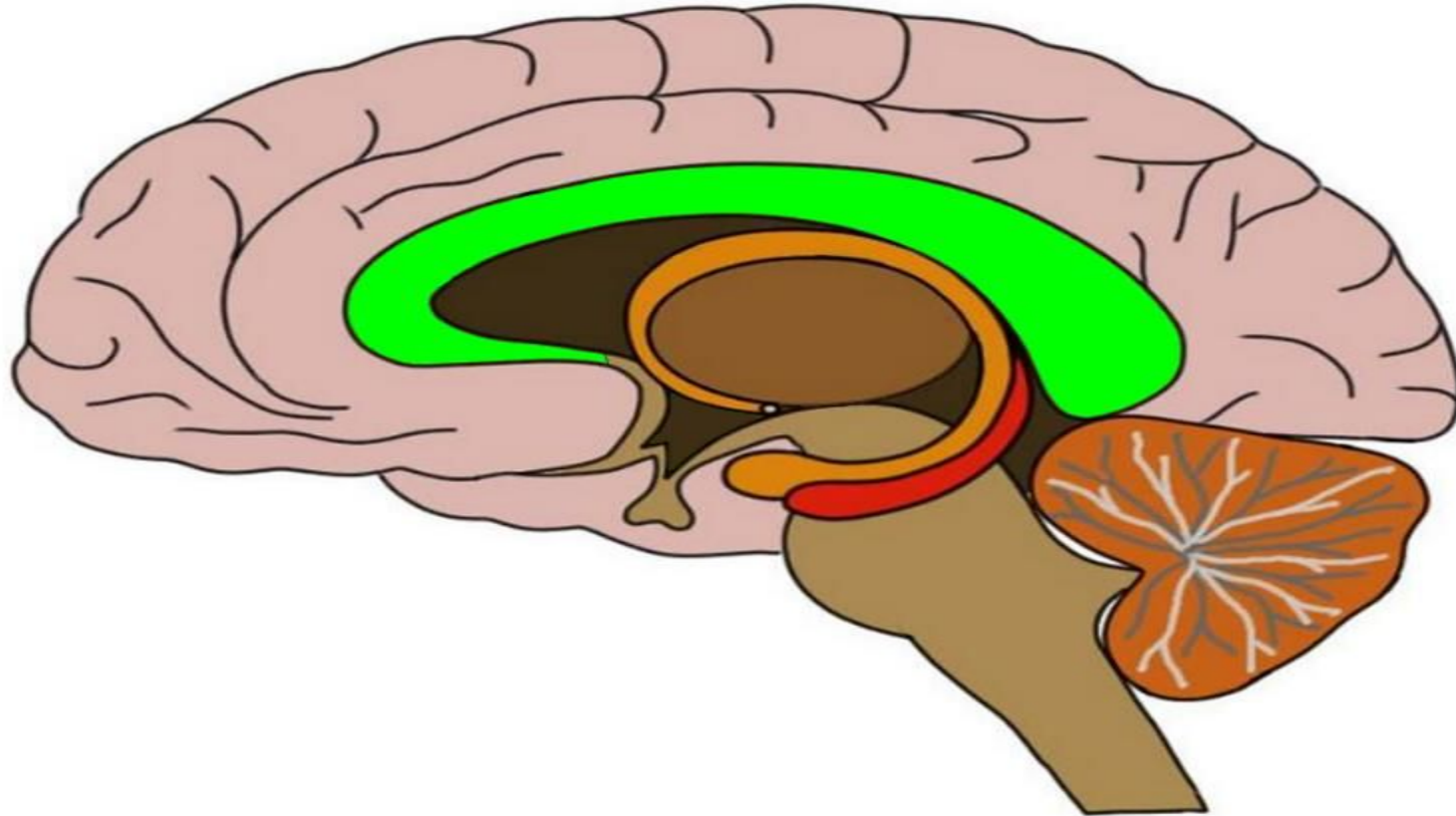


White Matter of Cerebral Hemispheres



Learning objectives:

At the end of the lecture, the students should be able to define:

- **Define commissural fibers of cerebral hemispheres.**
- **Enlist the examples of commissural fibers.**
- **Describe corpus callosum & its parts.**
- **Define association fibers, &**
- **Enlist the examples of association fibers.**
- **Define projection fibers with examples.**
- **Describe internal capsule & its parts.**

Introduction:

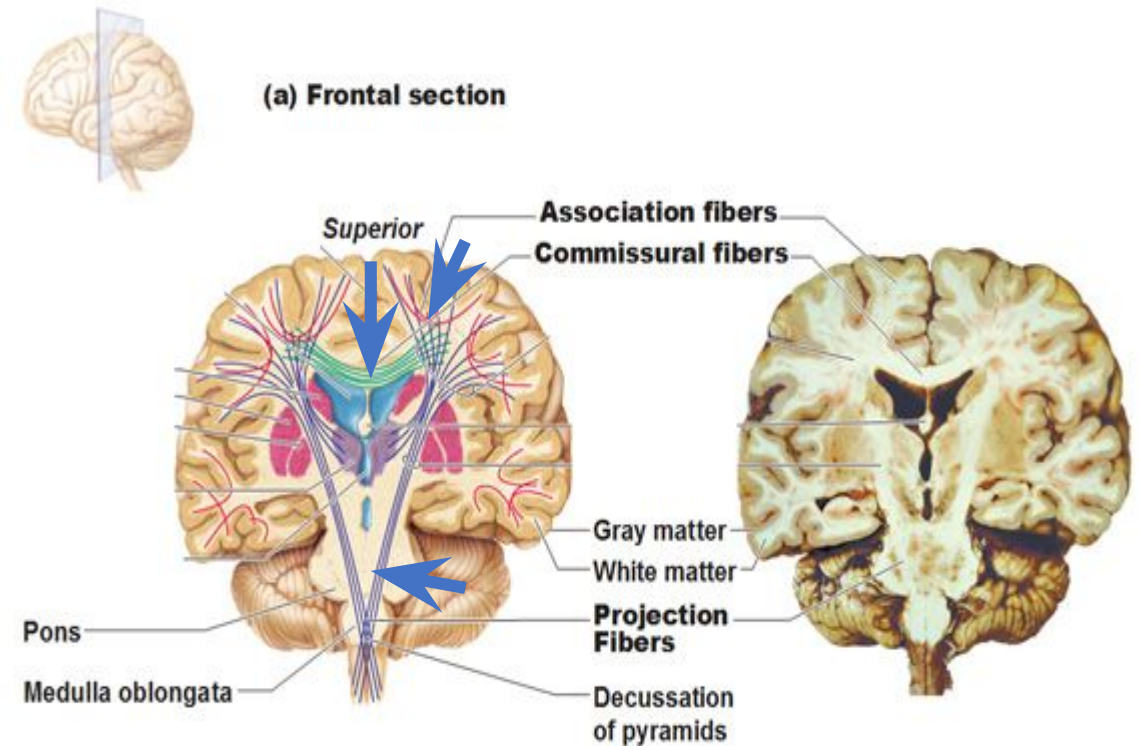
White matter is composed of myelinated nerve fibers supported by neuroglia.

Classification:

The nerve fibers (of white matter) can be classified into three groups according to their connections:

- 1) Commissural fibers
- 2) Association fibers, &
- 3) Projection fibers

Cerebral White Matter – 3 types of fibers



Commissural fibers:

Commissural fibers essentially connect the corresponding regions of the two hemispheres..

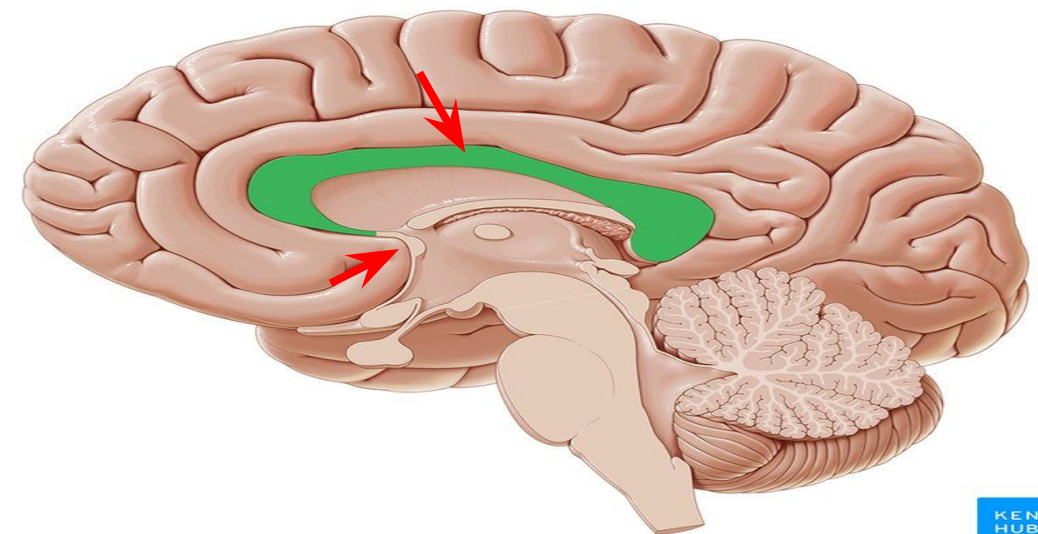
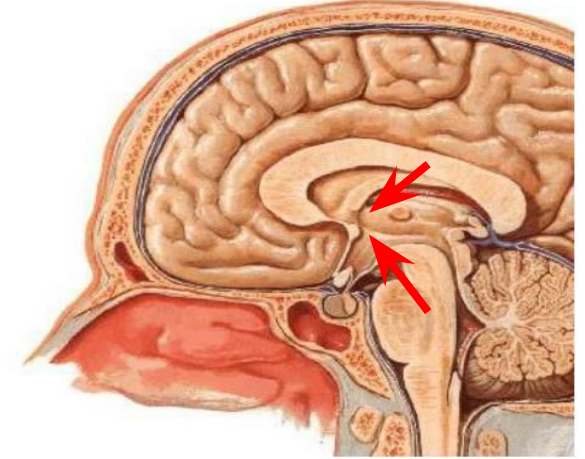
Examples: corpus callosum, anterior commissure, posterior commissure and habenular commissure.

Corpus callosum:

- It is the largest commissure which connects two cerebral hemispheres of brain.
- It lies beneath longitudinal fissure.

Commissural Fibers

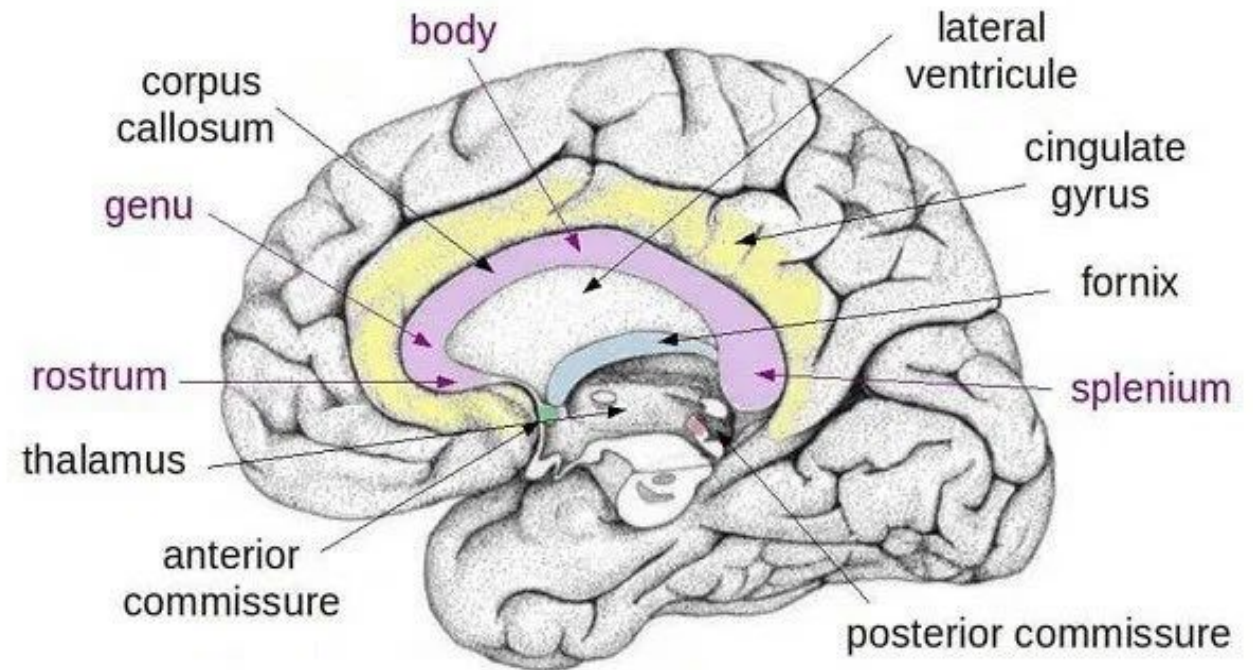
- Anterior Commissure
 - Inf. & mid. temporal gyri, olfactory areas
- Posterior Commissure
 - Preoptic nuclei (vision)
- Habenular Commissure
 - Habenular nuclei (olfaction)
- Corpus Callosum
 - Connects hemispheres
 - Rostrum, genu, body & splenium



Parts of corpus callosum:

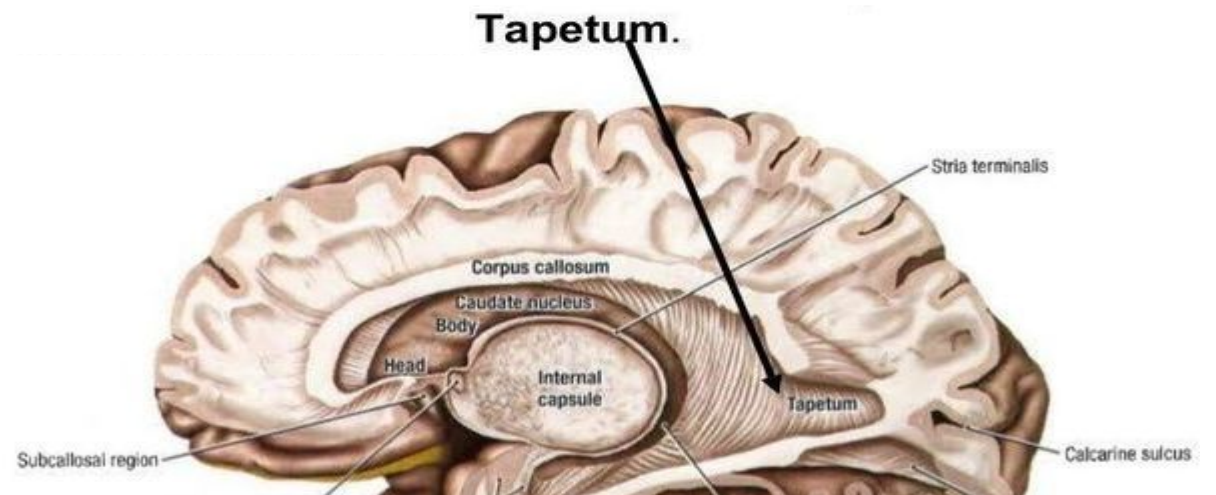
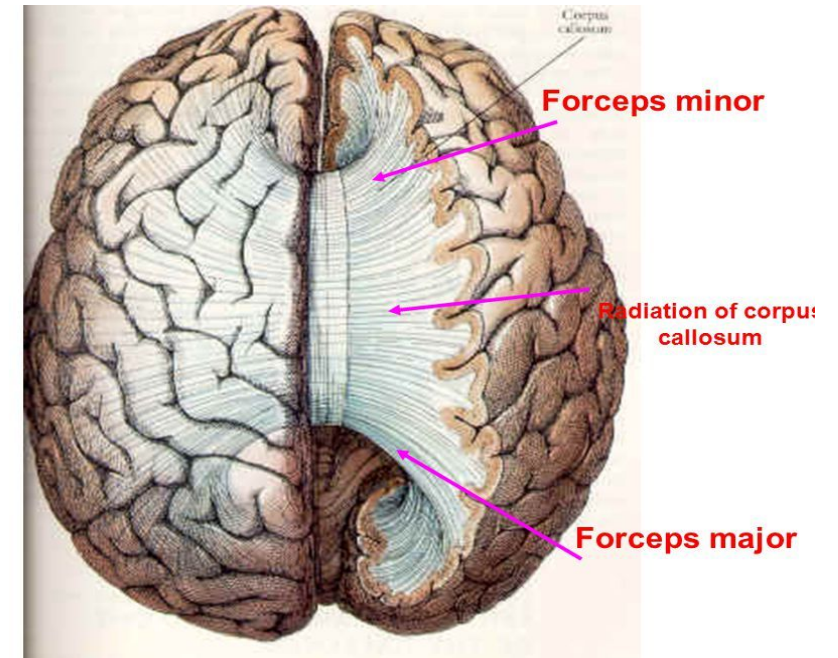
Corpus callosum is divided into:

- I. **Rostrum:** a thin part at the anterior end of corpus callosum.
- II. **Genu:** curved anterior end of corpus callosum.
- III. **Body:** long narrow portion that arches posteriorly, &
- IV. **Splenium:** terminal thick posterior portion.

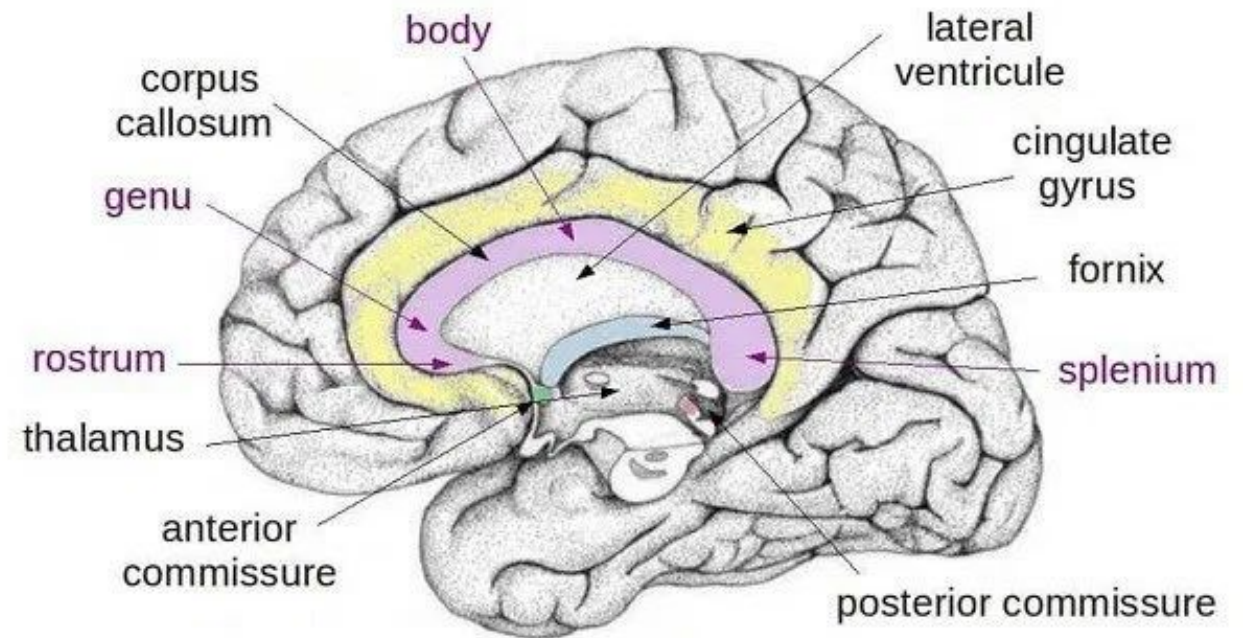
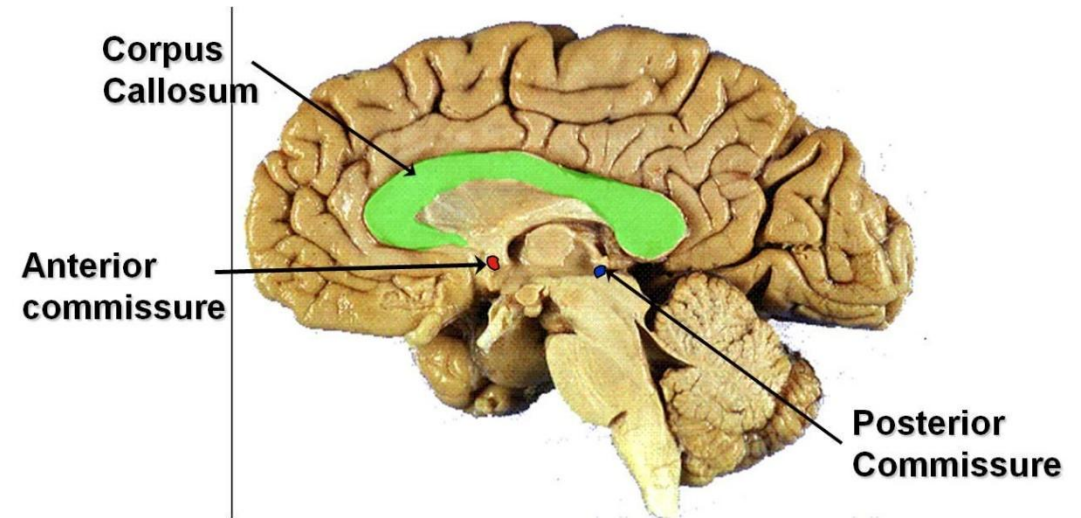


Terms related to corpus callosum:

- **Forceps minor:** fibers of genu which curve forward & connect anterior regions of the two frontal lobes.
- **Radiation of corpus callosum:** formed by the fibers of body which extend laterally & intersects with bundles of association & projection fibers.
- **Tapetum:** some fibers from radiation of corpus callosum which form the roof & lateral wall of posterior horn and lateral wall of inferior horn of lateral ventricle.
- **Forceps major:** fibers of splenium which curve backward & connect the posterior regions of the two occipital lobes.



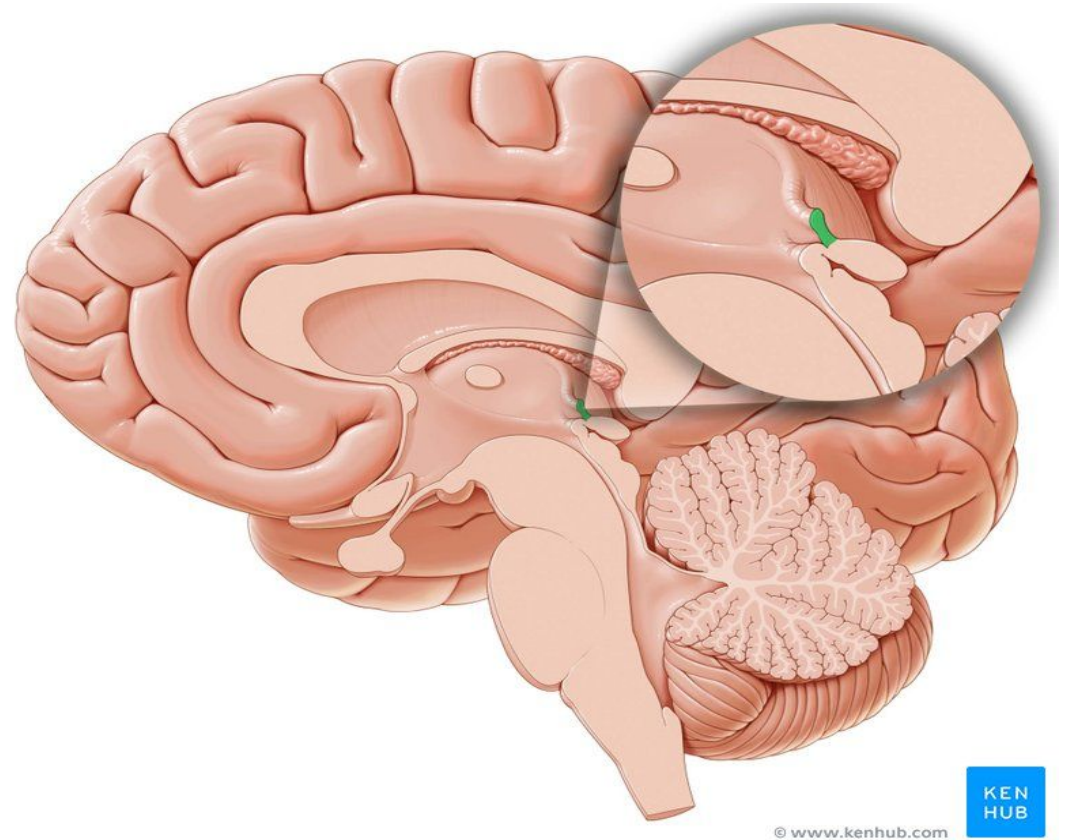
- **Anterior commissure:** bundle of nerve fibers which connects two olfactory bulbs and the temporal lobes of opposite cerebral hemispheres.
- **Posterior commissure:** is related to lower part of stalk of pineal gland & the fibers from pretectal nuclei involved in pupillary light reflex cross this commissure.
- **Fornix:** it constitutes efferent system of hippocampus & composed of fibers passing from mamillary bodies to hypothalamus.



Habenular Commissure:

A small bundle of nerve fibers that is located in the superior part of root of pineal stalk and connects the habenular nuclei of opposite sides.

- Habenular nuclei receive afferents from amygdala & hippocampus.
- These afferent fibers pass to habenular nuclei via **stria medullaris thalami**.



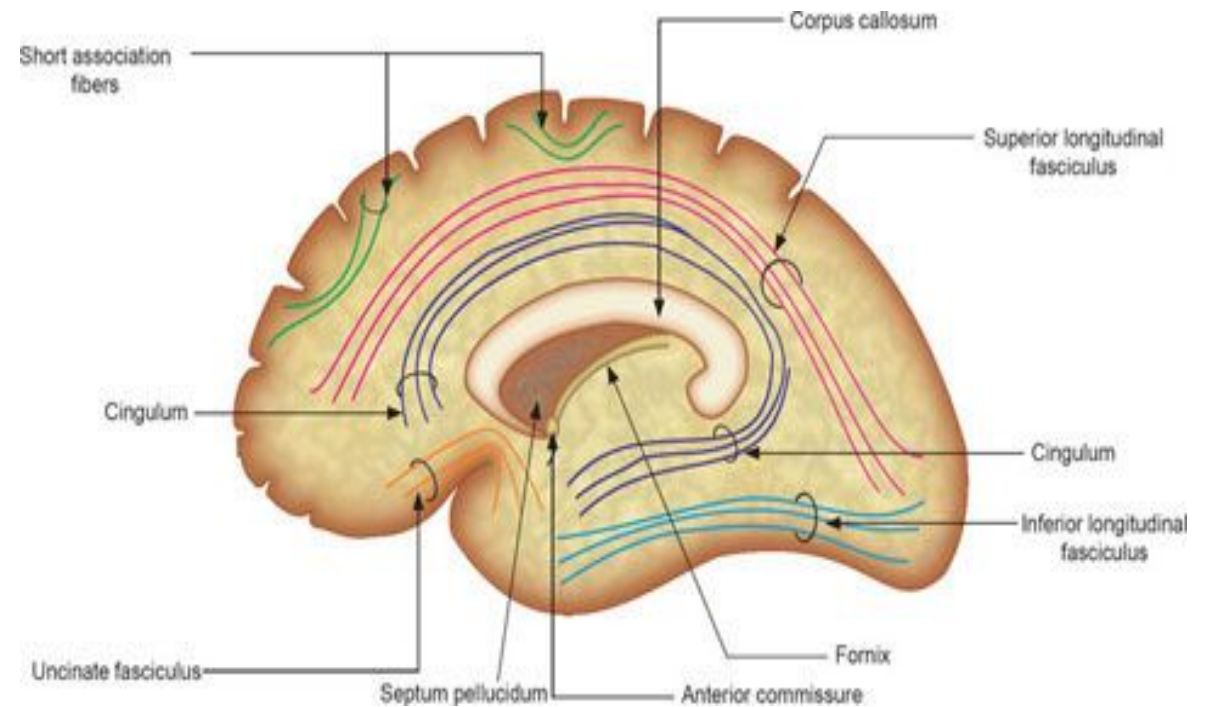
Association Fibers:

Association fibers essentially connect various cortical regions within the same cerebral hemispheres.

Examples: uncinate fasciculus, cingulum, superior & inferior longitudinal fasciculi.

- Association fibers may be divided into:

- **Short association fibers:** lie beneath cerebral cortex & connect adjacent gyri.
- **Long association fibers:** collected in the form of large bundles.

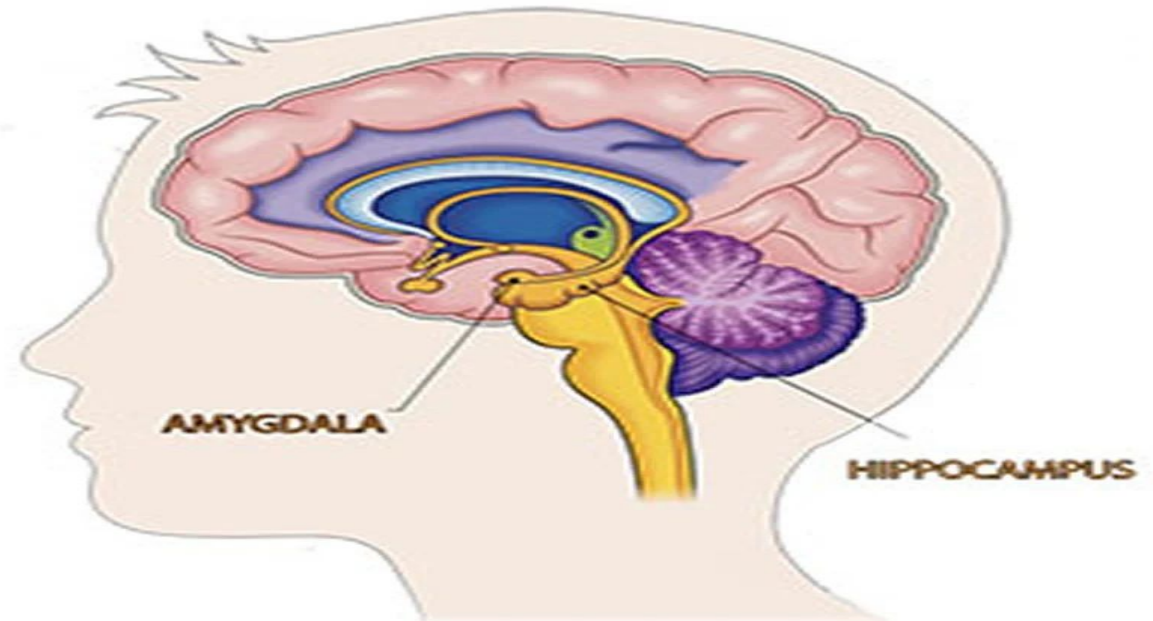
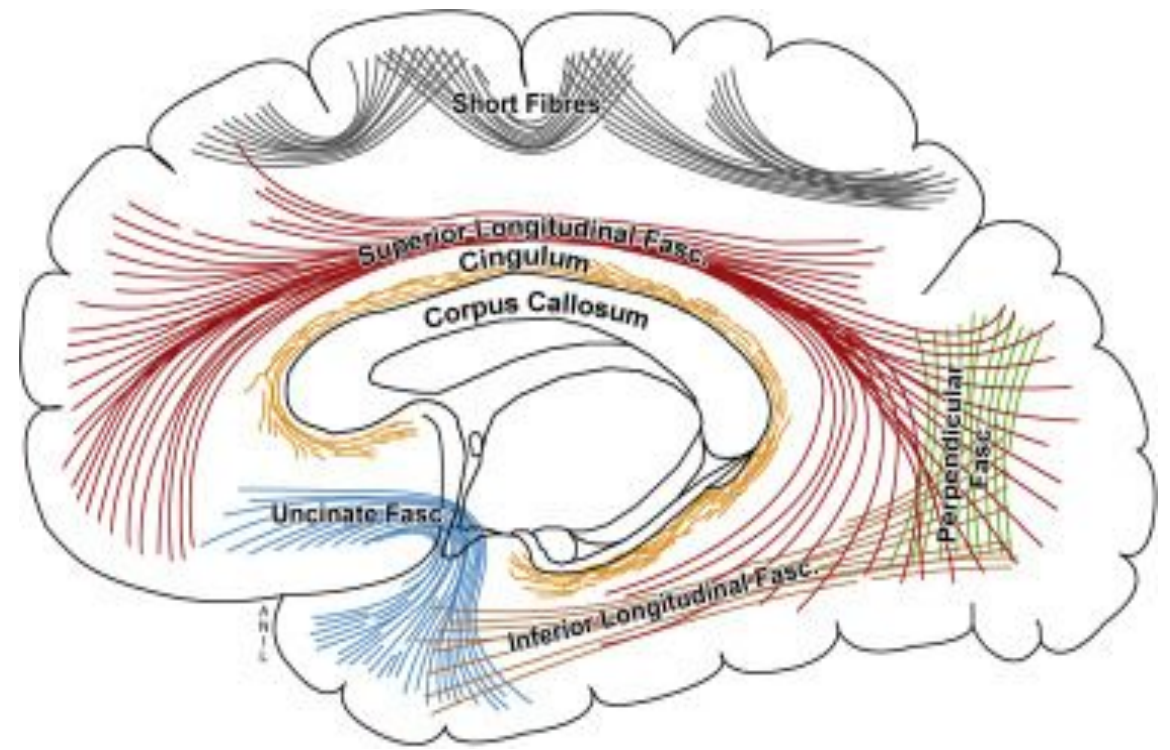


Uncinate fasciculus: Connects parts of the limbic system such as the hippocampus and amygdala in the temporal lobe with frontal ones such as the orbitofrontal cortex.

Cingulum: located in cingulate gyrus and connects frontal, parietal and medial temporal regions.

Superior longitudinal fasciculus: largest, connects anterior part of frontal lobe to occipital & temporal lobes.

Inferior longitudinal fasciculus: connects the occipital and temporal-occipital parts of the brain to the anterior temporal parts.



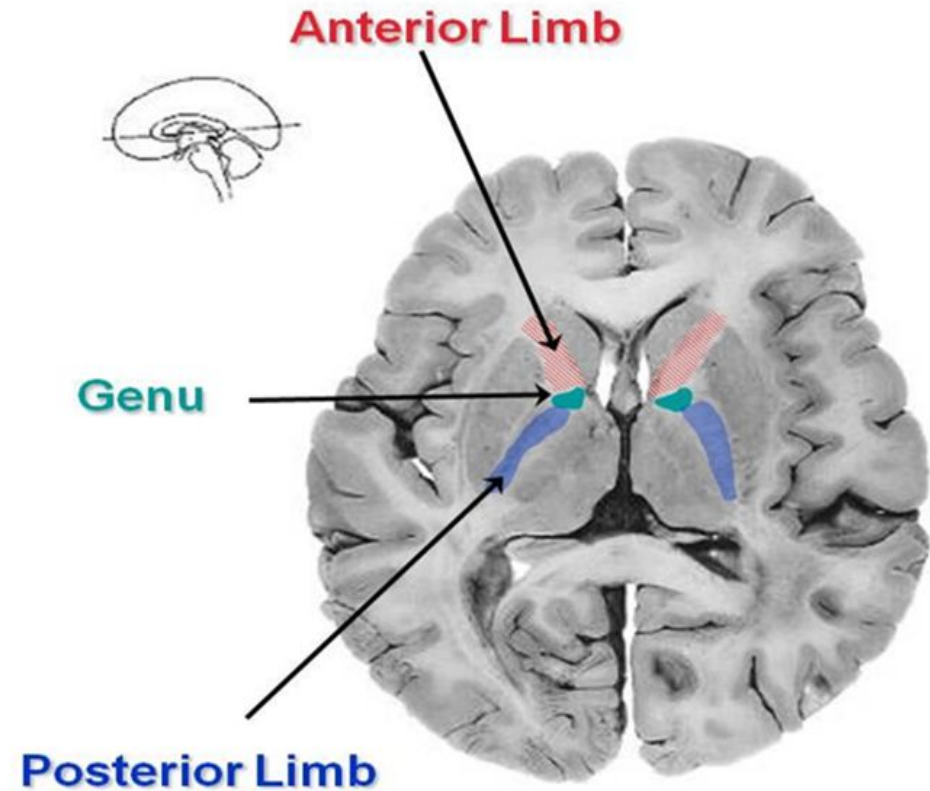
Projection fibers:

Are afferent and efferent nerve fibers which pass to & from the brainstem to the whole cerebral cortex and travel between large nuclear masses of gray matter within cerebral hemisphere.

Examples: internal capsule, corona radiata.

Internal capsule: it is a compact band consisting of ascending and descending fiber tracts which connect the cerebral cortex with the brainstem, and the spinal cord.

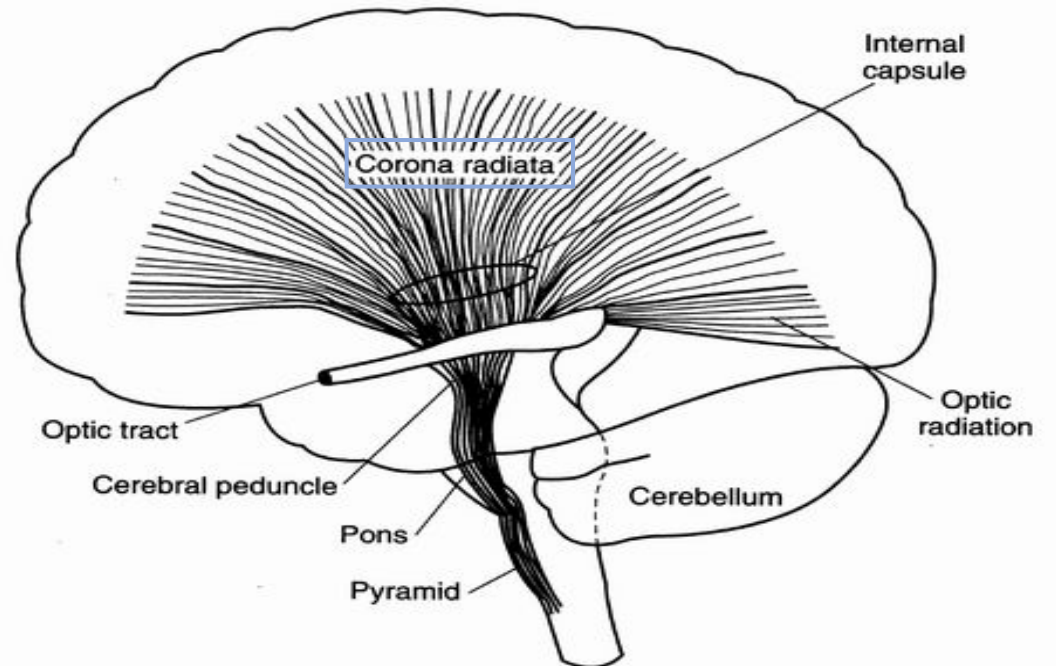
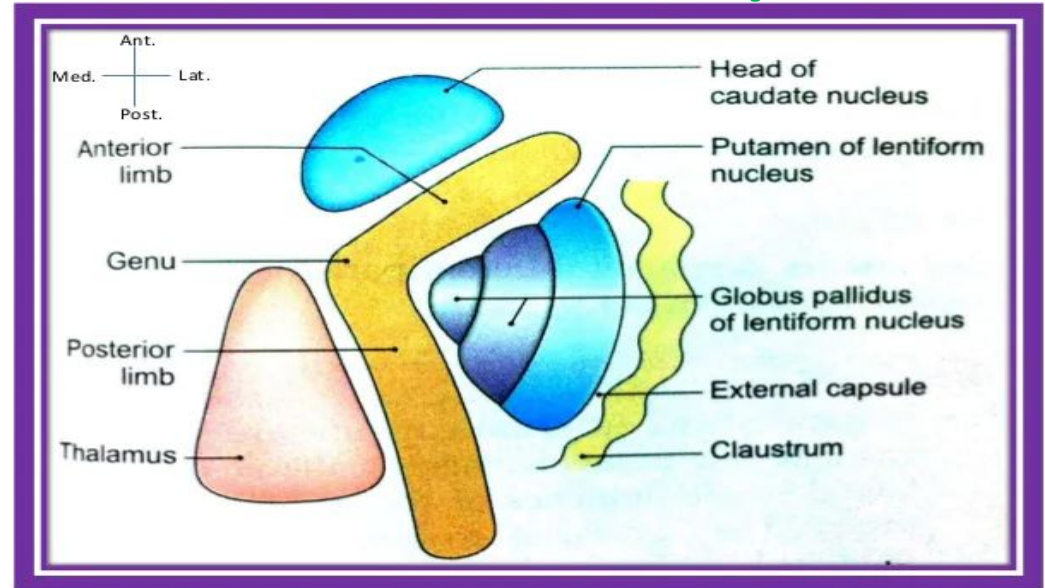
The Internal Capsule



- Internal capsule is bounded medially by caudate nucleus & thalamus and laterally by lentiform nucleus.
- Internal capsule has a **anterior limb**, a **posterior limb** which unite together at **genu**.
- It also has **retrolenticular** & **sublenticular parts**.

Corona radiata: The fibers of the internal capsule radiate widely as they pass to and from the various parts of the cerebral cortex, forming the corona radiata.

Parts of Internal capsule



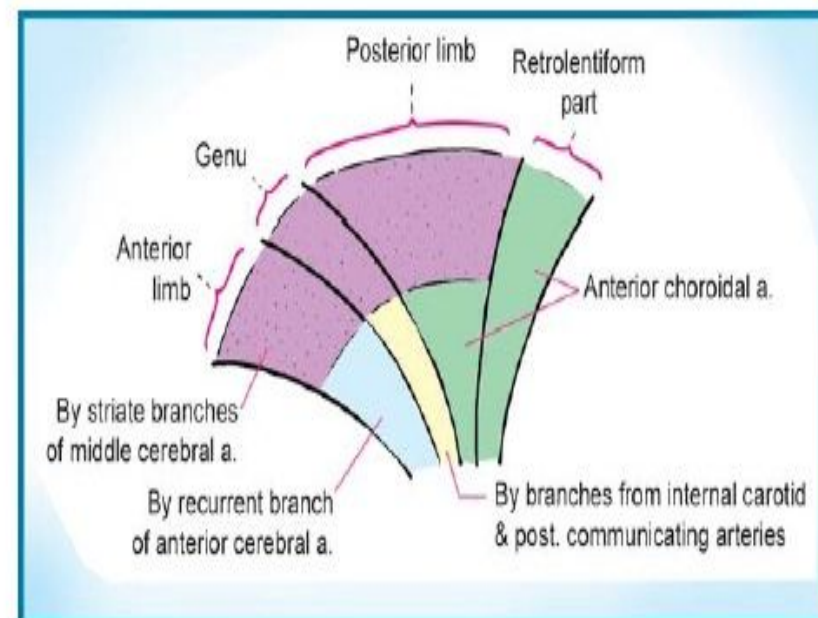
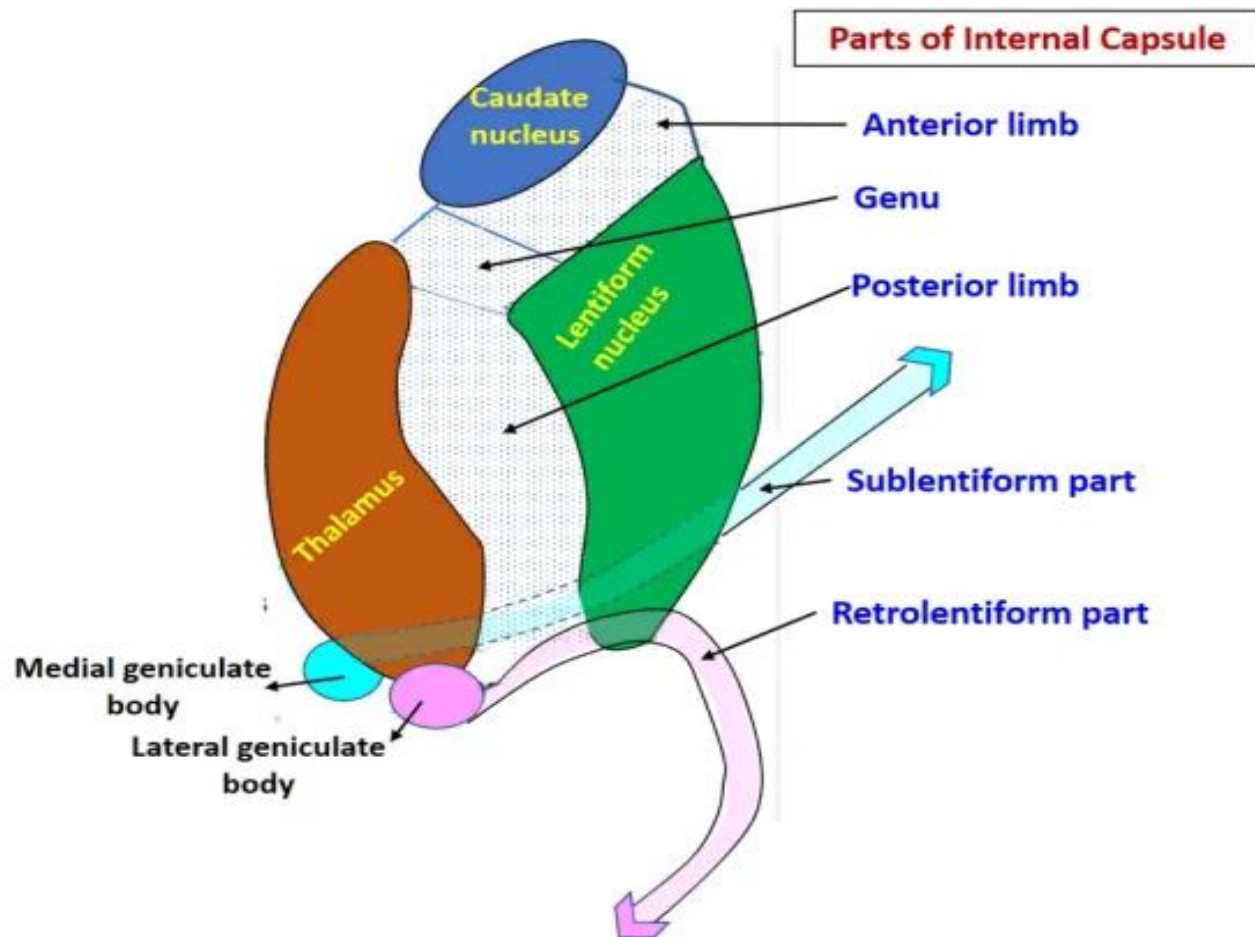
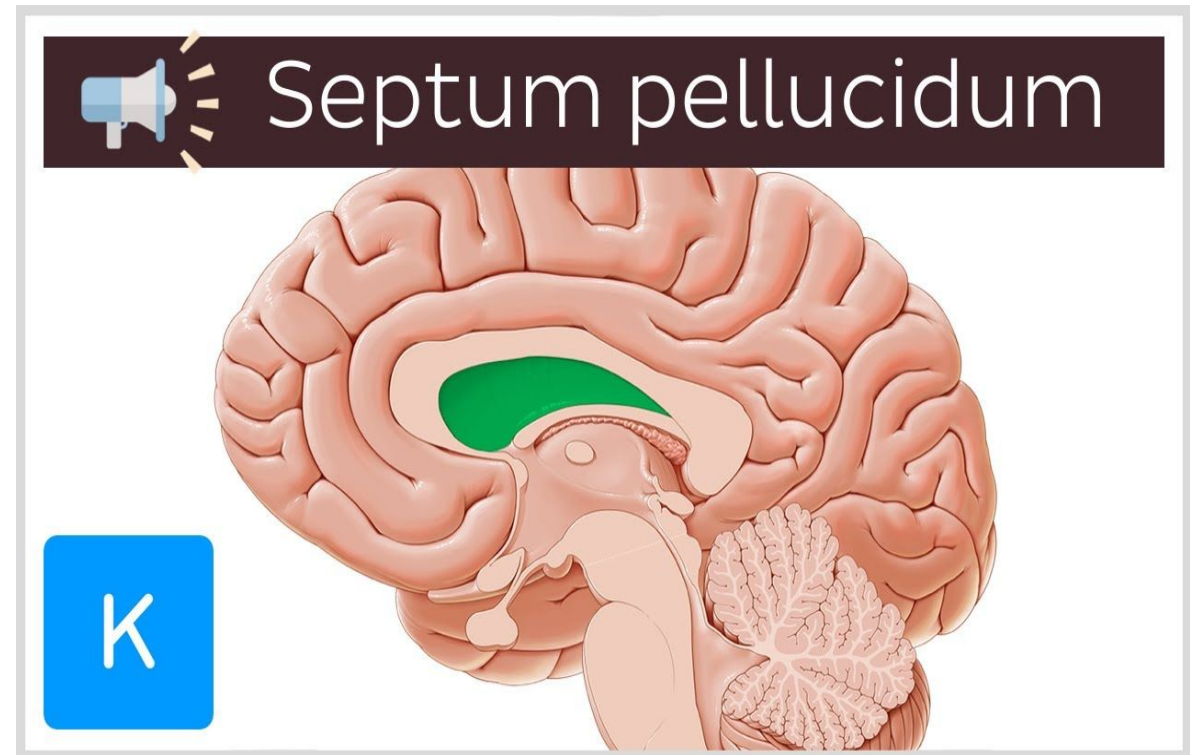


Fig. 21.8. Scheme to show the arterial supply of the internal capsule.

Septum pellucidum: a thin vertical sheet of nervous tissue containing white & gray matter covered on either side by ependyma, extends between fornix & corpus callosum.

- It is a double membrane with a closed slit-like cavity and forms a partition between anterior horns of lateral ventricles.



References:

- **CLINICAL NEUROANATOMY**

Richard S. Snell

8th edition

Chapter 7

Pages # 265-268